AI-Driven Intraday Trading Architecture

Consolidated design based on our discussion

Objective

Maximize risk-adjusted profit rather than prediction accuracy. A 54% accurate model can be profitable if expectancy is positive.

Layer 1: Forecast Models

R5, R10, and R15 represent expected returns over the next 5, 10, and 15 minutes respectively. Prefer return prediction over raw price prediction.

Layer 2: Confidence Models

For each horizon generate both expected return and confidence: R5/C5, R10/C10, R15/C15.

Layer 3: Risk Models

Forecast volatility and expected downside (Maximum Adverse Excursion). Consider reward-to-risk before trading.

Layer 4: Market Regime Model

Classify conditions such as Trending Up, Trending Down, Range Bound, High Volatility, or Low Volatility.

Layer 5: Meta-Labeling Model

Use outputs from return forecasts, confidence scores, volatility, regime, volume, spread and technical features to predict probability that a trade will be profitable.

Layer 6: Position Sizing

Position size should scale with confidence and trade quality rather than using a fixed size.

Trade Selection Logic

Trade when multiple horizons agree, confidence is above threshold, and the meta-model indicates high probability of profitability.

Key Metrics

Focus on Profit Factor, Sharpe Ratio, Sortino Ratio, Max Drawdown, and Expectancy rather than RMSE, MAE, or Accuracy alone.

Recommended Architecture

Forecast Models -> Confidence Models -> Volatility Forecast -> Regime Model -> Meta-Labeling Model -> Position Sizing -> Execution -> Risk Management.

Important Definitions

R5 = Expected return over next 5 minutes. R10 = Expected return over next 10 minutes. R15 = Expected return over next 15 minutes.

The most valuable next step is often not improving price prediction accuracy, but building a meta-labeling model that learns when the forecast models are likely to be correct.